

Edeon — Registration Readiness Dossier

Compound: Chlorantraniliprole (CAS: 500008-45-7)

SMILES: Cc1c(C(=O)Nc2cc(Cl)cc(C(=O)NC(C)C)c2Cl)nn(-c3c(Cl)cccn3)c1Br

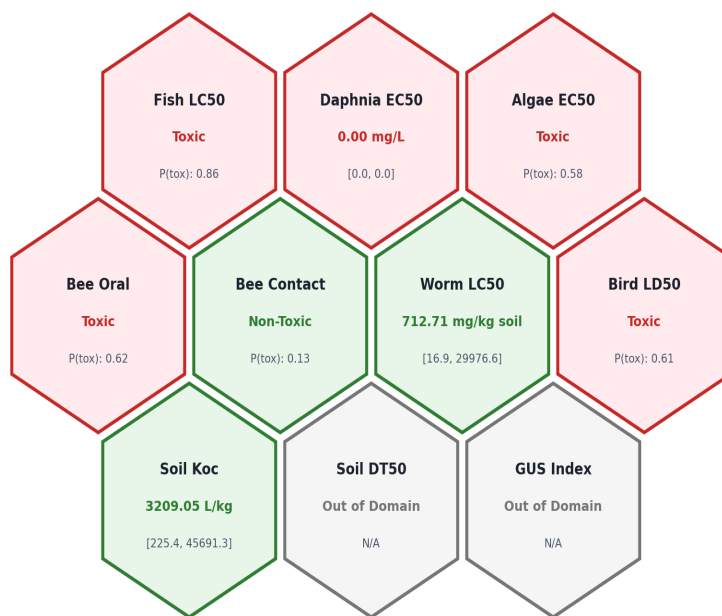
REGISTRATION READINESS VERDICT: NO_GO

Driver: Showstopper triggered: PBT/vPvB (REACH Annex XIII), Aquatic hazard (CLP H400/H410/H411/H412), Acute mammalian toxicity (CLP oral), Soil persistence (EU 1107/2009) | **Confidence:** high

Rationale: Soil DT50 = 1358 d > 180 d (vP), BCF = 13693 > 5000 (vB), Daphnia EC50 = 0.0000 mg/L < 0.1 (T); Most sensitive endpoint: Daphnia EC50 = 0.0000 mg/L, BCF = 13693 ≥ 500 (bioaccumulative), Not rapidly degradable (default assumption); Predicted rat oral LD50 = 3 mg/kg; Predicted soil DT50 = 1358 days, DT50 > 365 d — very persistent, likely to accumulate

Tier-1 Predictive Honeycomb

Chlorantraniliprole — QSAR Safety Profile



Regulatory Cut-Offs & Criteria Evaluation

Criterion / Endpoint	Status	Evidence & Details
Genotoxicity / Mutagenicity alerts (Benigni-Bossa)	PASS	No genotoxicity structural alerts detected
Endocrine disruption screening (structural)	PASS	No endocrine disruption structural alerts detected
Skin sensitization alerts (protein reactivity)	PASS	No skin sensitization alerts detected
PBT/vPvB (REACH Annex XIII)	LIKELY_SHO WSTOPPER	Soil DT50 = 1358 d > 180 d (vP); BCF = 13693 > 5000 (vB); Daphnia EC50 = 0.0000 mg/L < 0.1 (T)
Groundwater concern (EU DWD 0.1 µg/L trigger)	PASS	GUS = 1.55 < 1.8 (non-leacher); GUS classification: non-leacher
Aquatic hazard (CLP H400/H410/H411/H412)	LIKELY_SHO WSTOPPER	Most sensitive endpoint: Daphnia EC50 = 0.0000 mg/L; BCF = 13693 ≥ 500 (bioaccumulative); Not rapidly degradable (default assumption)
Acute mammalian toxicity (CLP oral)	LIKELY_SHO WSTOPPER	Predicted rat oral LD50 = 3 mg/kg
Soil persistence (EU 1107/2009)	LIKELY_SHO WSTOPPER	Predicted soil DT50 = 1358 days; DT50 > 365 d — very persistent, likely to accumulate

IN-SILICO SCREENING ONLY — These results are computational triage signals based on predicted endpoints and structural pattern matching. They are NOT regulatory determinations and cannot replace experimental studies or expert regulatory assessment.